

Cardinality Degradation in Structured Extraction: A Constructed-Document Pilot of Constrained and Unconstrained Decoding

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Abstract

Schema-constrained decoding is widely believed to cost accuracy, because forcing output through a grammar masks tokens the model would otherwise emit. We measure that cost along an axis it has not been measured on: the number of items the model must produce. Holding input length constant by construction, we build documents containing exactly k verbatim-validated sentences, each naming one entity, and vary k from 1 to 16. Extraction recall declines in k under unconstrained decoding, by 0.214 from $k \leq 3$ to $k \geq 16$ (Spearman -0.293 , $p = 0.011$), and does not decline under a JSON schema (-0.114 , i.e. recall rises; $p = 0.73$). Paired on the same 75 documents, 25 per level, unconstrained decoding leads by 0.200 at $k = 1$ and trails by 0.128 at $k = 16$; the gap moves by -0.082 per doubling of k ($p = 0.006$). The relative result is driven largely by the constrained arm’s behaviour at $k = 1$, where it emits an empty list on 13 of 25 documents; with those documents removed the interaction vanishes ($p = 0.79$), and from $k = 4$ to $k = 16$ both arms decline by a similar amount. At $k = 16$ the constrained arm emits 9.4 of 16 available items and the unconstrained arm 7.2, at equal precision, so the arms differ in how many items they stop at rather than in which items they find; why the schema changes the stopping point is not established here. The study’s pre-registered validity gate, recall of 0.80 at $k = 1$, was never met, so absolute recall levels are not interpretable and only the between-arm contrast is reported. We also show that item count, item density and document length cannot be held fixed simultaneously, which determines whether any measurement on this axis is an upper or a lower bound.

1 Introduction

Ask a language model for a list and it will sometimes give you a shorter one than the document supports. Whether that tendency worsens as the list grows is an empirical question with a measurable answer, and it has not been measured in the setting where it matters most for production systems: structured extraction under a schema.

Two literatures bracket the question without answering it. Long-context work measures how retrieval degrades with input length and with the position of the evidence [9, 6, 5], and multi-evidence variants report that retrieval falls as the number of inserted facts rises [7, 12]. Long-list extraction is recognised as hard enough to need retrieval augmentation and a pruning stage [11],

and extraction benchmarks report performance falling as schemas widen [2]. Structured-output work measures the cost of constrained decoding. The grammar masks tokens the model would otherwise emit [15]; reported accuracy losses of several points on reasoning tasks [14] are contested as artefacts of prompt mismatch [16] or of vocabulary misalignment [1]; grammar constraints help on structured NLP tasks without finetuning [3]; and a benchmark over ten thousand schemas measures validity, coverage and efficiency [4]. Closer to the present axis, Levy et al. [8] show that reasoning degrades as the same task is padded with more tokens, LongGenBench [13] measures how long structured *outputs* degrade with their required length, and NoLiMa [10] separates retrieval from lexical matching in long contexts. None of these varies the number of items to emit while holding input length fixed, and none compares decoding regimes on that axis.

The distinction matters because the two candidate explanations make opposite predictions. If degradation is a property of the model’s attention over a long list of targets, a grammar should not help. If it is a property of the model choosing when to stop, a decoding regime that changes the stopping point should help. The data are consistent with the second, with the qualifications in Section 5.

Contributions. We fix k , the number of items to emit, as an evaluation axis and hold input length constant by constructing documents rather than annotating them, which makes k exact and the gold set free. The result is reported as a slope, not an existence claim: the pre-registered decline is present under unconstrained decoding and absent under a schema, inverting the expectation that grammars cost accuracy, though the interaction between the two arms rests mostly on the $k = 1$ level and is reported with that caveat. A constraint that any study on this axis inherits is made explicit. Count, density and length are mechanically linked, so fixing two determines the third and decides the direction of the residual bias.

2 Constructing documents with known k

Annotation is the obvious approach and the wrong one. If k is a property of whichever documents happened to be sampled and a human must recover it, k is measured with error that correlates with the outcome, since an annotator’s attention flags on exactly the long enumerations where the effect should live.

We construct instead. The item bank is drawn from SEC 10-K filings: sentences each naming exactly one organisation, each verified to appear verbatim in its source document. Filler is drawn from the same corpus with every sentence containing a capitalised run removed, so it cannot carry a legitimate item. A document is k item sentences injected at uniformly random positions into filler, padded to a constant 9,000 characters. Gold is the injected set.

Three properties follow: k is exact, input length is constant by construction rather than merely uncorrelated, and position is averaged over rather than confounded with k .

The item bank is screened by measurement, not by rule. Three syntactic filters for “which sentences transplant into a synthetic document” were tried and all three failed, leaving recall at $k = 1$ far below any usable baseline. The bank is therefore defined empirically: a candidate sentence is retained only if the extractor recovers it at $k = 1$ in a short document. 143 of 240 candidates survive.

A validity gate, which was not met. The pre-registration (docs/HYPOTHESIS_CARDINALITY.md, amendment of 2026-09-10) requires recall at $k = 1$ to clear 0.80 before any slope from a cell is interpretable: one item, one document, a sentence the extractor itself produced in the wild. The gate failed three times during construction (0.333, 0.400, 0.400), and the pre-registration’s stop rule states that if it failed again after one further rebuild, the constructed design would be abandoned and the study written up as a dilution measurement and a negative report. The rebuild raised $k = 1$ recall to 0.440 under constrained decoding and 0.640 under unconstrained decoding. Neither clears 0.80. The stop rule therefore called for abandoning the design, and we proceeded anyway. The consequence is stated here and carried through the paper: absolute recall levels in this study are not interpretable as properties of the extractor, and only the contrast between the two decoding arms on identical documents is reported. The three failures were nonetheless informative. The decisive diagnosis came from holding the item and k fixed and varying only the filler:

filler	recall at $k = 1$
1,500 characters	0.667
4,000 characters	0.583
14,000 characters	0.417

Dilution alone is worth 0.25 recall. The filler had been selected *for containing no entity names*, which made it far more inert than any real prompt, where a passage selector retains only relationship-dense text. Document length was reduced accordingly.

3 The count–density–length trilemma

The dilution measurement exposes a constraint that binds every study of this axis. Item density is mechanically determined by the other two quantities: $\text{density} = (k \times \text{span length}) / \text{document length}$. Fixing any two fixes the third.

Held fixed	Varies	Residual confound
length and count (this work)	density	density aids high k ; effect is a lower bound
density and count	length	length suppresses high k ; an upper bound
length and density	—	k cannot vary

We hold length fixed, so item density rises with k : a $k = 16$ document is denser in items than a $k = 1$ document of the same length. Since dilution demonstrably suppresses recall, high- k documents receive a density advantage that works against the hypothesis. Our estimate is therefore conservative.

4 Design and pre-registration

The pre-registration is a dated, amended document (docs/HYPOTHESIS_CARDINALITY.md) and the history matters for reading the result. The hypothesis, that recall declines in k at fixed input length, and the rejection rule, which requires *both* a recall drop of at least 0.15 from $k \leq 3$ to $k \geq 16$ and $p < 0.05$ on the rank test, were fixed before any cell was run and did not change. The second factor did. The original design was a 2×2 of model size (qwen3:32b against

qwen3:30b-a3b) by reasoning mode (think against no-think). On 2026-09-11 the reasoning-mode cells came back identical to three decimals, because on this stack the flag only changes which response field the same tokens are filed under, and the factor was replaced by constrained against unconstrained decoding. That replacement was recorded before the unconstrained arm was run, and the literature review had already named decoding regime as the strongest competing explanation, but the decoding comparison was not part of the original pre-registration and the interaction test in Section 5 was not pre-specified at all. The repository’s first commit (2026-09-14) contains both the pre-registration and the results, so the ordering rests on the document’s own dated amendments rather than on commit history. Bucket means are for display; estimation is per document.

Both arms use qwen3:14b at temperature 0 with a 32,768-token context and a 24,000-token generation ceiling, on the same 75 documents, 25 at each of $k \in \{1, 4, 16\}$. The constrained arm is parsed with the strict schema parser, since its output is grammar-guaranteed to validate. The unconstrained arm is prompted for JSON and parsed post hoc with a deliberately tolerant parser that strips reasoning blocks and code fences and scans for the last balanced object: a parse failure would score as a recall failure and confound the arm under test. The asymmetry favours the unconstrained arm, if anything.

The ceiling control matters and is easy to get wrong. An earlier attempt used a 30B mixture-of-experts model, which writes 7,000 to 12,000 tokens unconstrained against an 8,192-token window. Four of six $k = 16$ documents truncated mid-array, parsed to zero items, and would have reported constrained decoding as hugely superior at the level that decides the study. That arm was discarded rather than repaired. In the reported runs, zero documents truncate and the maximum generation is 686 tokens.

5 Results

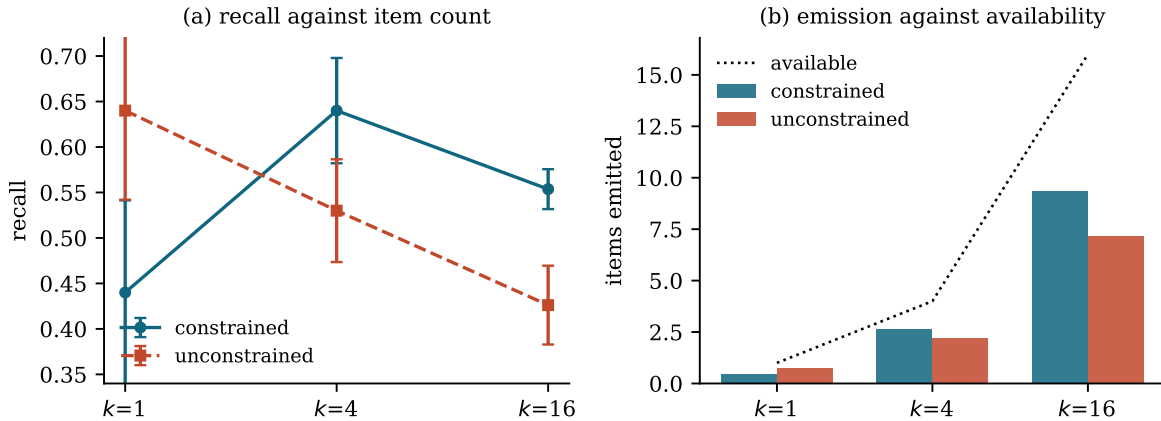


Figure 1: (a) Recall against item count for both decoding regimes, 75 matched documents per arm, error bars one standard error. The arms cross between $k = 1$ and $k = 4$. (b) Items emitted against items available. The constrained arm tracks availability more closely as k grows; the unconstrained arm falls away.

k	constrained	unconstrained	paired difference	p
1	0.440	0.640	+0.200	0.096
4	0.640	0.530	-0.110	0.110
16	0.554	0.426	-0.128	0.005

Table 1: Recall by item count and decoding regime, paired on identical documents.

The pre-registered test rejects for one arm and not the other. Unconstrained decoding drops 0.214 from $k \leq 3$ to $k \geq 16$ with Spearman -0.293 , $p = 0.011$, satisfying both conditions. Constrained decoding drops -0.114 , meaning recall rises, with Spearman $+0.041$, $p = 0.73$, satisfying neither.

The $k = 1$ comparison favours unconstrained decoding by 0.200 at $p = 0.096$. Per-document recall at $k = 1$ is close to a Bernoulli draw, and the variance arithmetic implies 106 documents are needed to detect a 0.15 difference at 80% power against the 25 run.

The interaction test. The pre-registered test is applied to each arm separately and never tests the claim the paper actually makes, which is that the slope differs between arms. That test was added after the data were in and is reported as such (`scripts/interaction_test.py`). Regressing the paired difference, unconstrained minus constrained, on $\log_2 k$ gives a slope of -0.082 per doubling (s.e. 0.029, $p = 0.006$); Spearman of the paired difference against k is -0.403 ($p = 0.0003$). The relative claim survives. It does not survive intact. At $k = 1$ the constrained arm returns an empty list on 13 of 25 documents, against 8 of 25 unconstrained, and those 13 documents carry the interaction: with them removed the slope is $+0.007$ ($p = 0.79$) and Spearman is -0.185 ($p = 0.15$). Within the unconstrained arm alone, recall from $k = 4$ to $k = 16$ falls from 0.530 to 0.426 with Spearman -0.242 ($p = 0.09$); within the constrained arm the same comparison falls from 0.640 to 0.554 with Spearman -0.237 ($p = 0.10$). Above $k = 1$ the two arms decline by a similar amount. The honest statement of the result is therefore narrower than the pre-registered verdicts suggest: the unconstrained arm meets the rejection rule and the constrained arm does not, but the difference between them is concentrated at the one level the study’s own power arithmetic and validity gate both flag as unreliable.

Emission counts. At $k = 16$ the constrained arm emits 9.4 of 16 available items and the unconstrained arm 7.2, at precision 0.946 and 0.948 respectively. The difference is not a precision–recall trade: the unconstrained arm produces fewer items, and relatively fewer as the list lengthens. Why the schema changes the stopping point is not established here. The schema is a bare `list[Link]` with no minimum length, so the grammar permits closing the array after any element, including after none, and the constrained arm does stop early, at 9.4 of 16, and emits an empty list on 13 of 25 documents at $k = 1$. An earlier draft attributed the difference to a partially filled array failing to parse; that is false for this schema. The manipulation that would test a grammar-level stopping account, a `minItems` bound or a required count field, was not run.

Cost. Unconstrained decoding is not cheaper. It costs $1.1\times$ the tokens at $k = 1$ and $0.7\times$ at $k = 16$: on this model it is both less accurate at high k and no less expensive.

6 Scope conditions

One model, dense, one family; nothing here separates architecture from decoding regime. The reasoning-mode flag on this stack changes only which response field the same tokens arrive in, verified by identical evaluation counts, so this work says nothing about inference-time reasoning. The item bank is drawn from a single domain and was screened with a different model (qwen3:30b-a3b) from the one evaluated (qwen3:14b), so the bank is not guaranteed recoverable at $k = 1$ by the evaluated extractor, which is one candidate explanation for the failed gate. Absolute recall levels are not interpretable, both for that reason and because the validity gate was not met. The between-arm contrast is the estimand, and it rests mostly on $k = 1$, the level that is underpowered by the study’s own arithmetic. Twenty-five documents per level and one model is a pilot scale; the next run needs at least 100 documents per level and more than one model family.

7 Conclusion

On the axis of how many items a model must emit, the grammar is not what costs recall. Under the pre-registered rule, unconstrained decoding shows a decline in k and schema-constrained decoding does not. The interaction between the arms is real on the full sample but is carried by the $k = 1$ level, where the constrained arm under-emits and the study’s validity gate was not met; above $k = 1$ the arms decline together. The arms differ in where the model stops rather than in what it retrieves, and the mechanism behind that difference is unresolved.

For practitioners the implication is narrower than the title suggests: on this model, free-form generation with post-hoc parsing, often proposed as an escape from grammar masking [14], does not recover the recall it gives up on long lists, and costs no fewer tokens. Whether a schema actively protects recall at high k is not settled by this study. For evaluation, a single F_1 over a benchmark whose k distribution is unstated is not comparable across benchmarks, because it averages over a mix nobody reported.

Reproduction. Code, pre-registration, the constructed evaluation set and all 450 raw model responses are released at <https://github.com/quantraunak/cardinality-eval>. The reported result re-scores without a GPU.

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